

END TERM EXAMINATION

THIRD SEMESTER [B.TECH] DECEMBER 2025 – JANUARY 2026

Paper Code: ES-201

Subject: Computational Methods

Time: 3 Hours

Maximum Marks: 60

Note: Attempt five questions in all including Q. No. 1 which is compulsory. Select one question from each unit. Scientific calculator is allowed.

Q. No.	Question (Attempt any four parts from Q1)	Marks
1(a)	Explain the rapid convergence of Newton-Raphson iteration by showing that the convergence is "quadratic".	[5]
1(b)	Use the Newton methods to Maximize: $z = -(x_1 - \sqrt{5})^2 - (x_2 - \pi)^2 - 10$ to within a tolerance of 0.05.	[5]
1(c)	Compute the integral $I = \int_{-2}^2 e^{-x^2} dx$ using 3-point Gaussian quadrature.	[5]
1(d)	Using the data $\sin(0.1) = 0.09983$ and $\sin(0.2) = 0.19867$, find an approximate value of $\sin(0.15)$ by Lagrange interpolation. Obtain the maximum truncation error bound.	[3+2]
1(e)	Estimate the computational efforts required in terms of arithmetic operations in Gauss elimination method to solve system of linear equations during elimination process and substitution process, and solve $3x_1 + 2x_2 = 18$, $-x_1 + 2x_2 = 2$ using Gauss elimination method.	[3+2]
1(f)	State whether the following piecewise polynomial is spline or not: $f(x) = \{ x^2 + 1, 0 < x < 1; 2x^2, 1 < x < 2; 5x - 2, 2 < x < 3 \}$	[5]
1(g)	Estimate the value of $y(0.20)$ by Picard's method, given that $dy/dx = x + y + xy$ and $y(0.0) = 1.00$ taking $h = 0.05$.	[5]
1(h)	Using Modified Euler's method, find $y(0.2)$ correct to four decimal points given that $y' = y + e^x$, $y(0) = 0$.	[5]
UNIT-I		
2(a)	Find the value of function $f(x) = 1 - \cos(x)$ at $x = 0.1$. Modify the expression so that loss of significant digits can be avoided and calculate the value again. Compare two values with the true value 0.4996×10^{-2} .	[6]
2(b)	Explain Lagrange's mean value theorem and verify the statement for the function $f(x) = x^3 + 2x$ in $[0, 3]$.	[1.5+2.5]
3	[OR] Estimate the minimum value of $f(x, y) = x^3 - 4x + y^2 - xy$ using Nelder-Mead method up to five iterations. Given initial vertices are $v_1 = (0, 0)$, $v_2 = (1.2, 0)$ and $v_3 = (0, 0.8)$.	[10]
UNIT-II		
4(a)	Prove that $\Delta^n \sin(ax + b) = [2 \sin(ah/2)]^n \sin[ax + b + n(ah + \pi)/2]$, h being the interval of differencing.	[6.5]
4(b)	For given $f(x) = 1/x^2$, find the divided differences $[a, b]$ and $[a, b, c]$.	[3.5]

[P.T.O.]

Q. No.	Question (Continued)	Marks								
5(a)	[OR] If the integral $\int_0^{\pi} e^{\cos x} dx$ is to be computed with absolute error less than $\frac{1}{2} \times 10^{-3}$, by composite trapezoidal rule with a uniform partition, how many subintervals are required to compute the value less than $\frac{1}{2} \times 10^{-3}$?	[6]								
5(b)	Use Romberg's method to estimate $\int_0^1 [1 / (1 + x^2)] dx$ correct to three decimal places taking $I(0.5) = 0.7084$, $I(0.25) = 0.6970$ and $I(0.125) = 0.6941$. Compare the results with the true value.	[3+1]								
UNIT-III										
6(a)	Test the consistency of the system of following given equations: $x_1 + 4x_2 + ax_3 = 6$, $2x_1 - x_2 + 2ax_3 = 3$, $ax_1 + 3x_2 + x_3 = 5$ Find the value of a so that system has unique solution, no solution and infinitely many solutions?	[5]								
6(b)	Determine the largest eigenvalue and corresponding eigenvector of the matrix: $A = [[2, -1, 0], [-1, 2, -1], [0, -1, 2]]$ using power series method taking starting eigenvector $x_0 = [1, 1, 1]^T$.	[5]								
7(a)	[OR] Obtain the linear spline for the function f(x) defined by the data given below: <table><tr><td>x</td><td>1</td><td>2</td><td>4</td></tr><tr><td>f(x)</td><td>1</td><td>5</td><td>11</td></tr></table> Evaluate the function at x = 3 .	x	1	2	4	f(x)	1	5	11	[5+1]
x	1	2	4							
f(x)	1	5	11							
7(b)										
UNIT-IV										
8(a)	Using the Taylor series for y(x) , compute y(0.1) correct to four decimal places if y(x) satisfies y' = x - y² and y(0) = 1 . Compare the result with the exact result value of y(0.1) .	[4+1]								
8(b)	Consider the initial value problem dx/dt = t + x² where x(0) = 1 . Using Runge-Kutta method of 4th order find x(0.2) .	[5]								
9	[OR] Apply Milne's method, to find the solution of the first order differential equation dy/dx = x - y² in the range 0 < x < 1 for the boundary conditions y = 0 at x = 0 taking h = 0.2 . Find first three required values y₁, y₂, y₃ and corresponding f₁, f₂, f₃ by Picard method.	[10]								

**** End of Question Paper ****